École polytechnique fédérale de Lausanne

Data Visualization

Milestone 1: The modern Olympics

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1 Dataset

The project uses a historical Olympics dataset sourced from Kaggle Griffin (2018a), covering all Olympic Games from Athens 1896 to Rio 2016. The data is organized at the athlete–event level, meaning each row represents one athlete competing in a specific event. The dataset contains 15 variables, including athlete characteristics (e.g., name, sex, age, height, weight), team and National Olympic Committee (NOC), sport, event, medal type, and contextual information such as year, season (Summer/Winter), and host city. In total, the dataset comprises 271,116 observations.

Overall data completeness is high, though some missing values are present. Approximately 23% of entries for height and weight are missing, and around 3% of age values are unavailable. Missing medal entries correspond primarily to athletes who did not win a medal. These are recoded as "No medal" for analytical clarity, acknowledging that a small fraction may reflect true missingness. All other variables have no missing data.

A review of dataset documentation and community discussions Griffin (2018b) highlights minor data quality issues. These include incomplete representation of athletes with non-ASCII characters in their names, occasional inconsistencies in extreme age values, and the assignment of athletes to modern countries rather than strictly historical entities. While such issues affect a small proportion of records, they are considered when interpreting results.

Preprocessing focuses mainly on ensuring consistency for analysis. In particular, medal counts are adjusted for team sports, where multiple athletes may be credited for a single medal; these are aggregated to reflect one medal per country. Additional transformations include filtering subsets of the data depending on the research question and merging with the `noc_country.csv` dataset to map NOCs to countries for geographical analysis. Overall, the dataset is well-suited for visualization with moderate preprocessing requirements.

2 Problematic

The main objective of this project is to build an interactive overview of the Olympic Games over time. Rather than relying on static summaries or single-point comparisons, the goal is to enable users to explore how participation, performance, and the structure of the Olympics evolve across more than a century. The project is motivated by the observation that many people engage with the Olympics during major events and are often curious to learn more about the sports and their history. An additional goal is to highlight lesser-known and historical sports disciplines to provide a richer perspective on the evolution of the Games.

The target audience includes casual sports fans seeking accessible and engaging insights, as well as history-oriented users interested in how historical events have shaped Olympic sports culture. The challenge is therefore to present a complex, multi-dimensional dataset in a way that remains intuitive while still conveying meaningful patterns.

To structure this exploration, the project focuses on three main axes:

- **Geographical perspective:** How do countries perform over time in terms of participation and success? This includes examining how global patterns shift across historical periods and how major geopolitical events (e.g., boycotts or the emergence and disappearance of nations) are reflected in Olympic participation. This perspective will be supported by interactive maps and timelines.
- **Athlete perspective:** How do athlete characteristics such as age, height, and weight vary across sports and over time? This perspective highlights differences between disciplines and provides insight into how athlete profiles evolve. Additionally, an interactive feature will allow users to input their own height and weight to identify sports that best match their profile.
- **Sports perspective:** How has the set of Olympic sports changed throughout history? This includes the introduction and removal of disciplines, illustrating how the structure of the Games adapts over time. This section will also incorporate links to external resources (e.g., Wikipedia) to encourage exploration of historical and less familiar sports.

3 Exploratory Data Analysis

The Exploratory Data Analysis of this aims to decode the complex interplay between athletic performance, socio-demographic shifts, and global geopolitical dynamics. Due to space limitations, not all exploratory data analysis is shown with figures; the corresponding results are provided in the supplementary notebook.

3.1 National Performance

By recalibrating the unit of analysis from the individual participant to the unique event, the study eliminates the data redundancy inherent in team sport records. This normalization protocol ensures that each collective victory is registered as a singular national achievement. We identified the consistent athletic proficiency of nations such as the United States, and the Soviet Union across multiple disciplines. The total count of medals per countrys over time are shown in Figure 1.

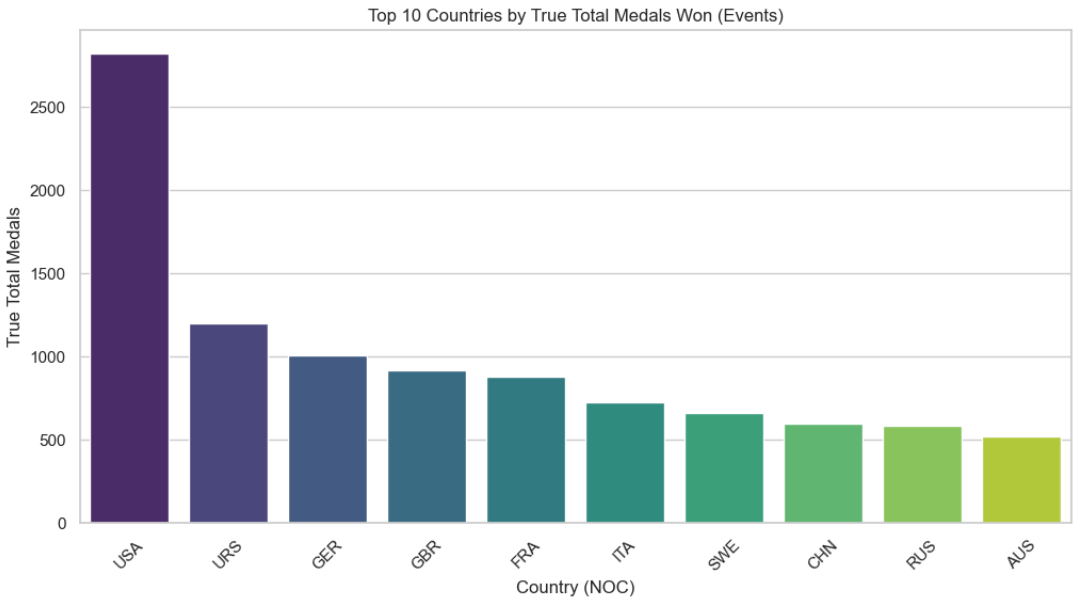

Figure 1: Top 10 countries (by National Olympic Committees, NOCs) with the highest total number of Olympic medals.

3.2 Trends in Gender Parity

The dataset provides a comprehensive record of the sociological evolution of gender inclusivity within the Olympic movement. By tracking unique participants categorized by sex, the analysis illustrates a clear trajectory from an almost exclusively male-dominated inception to the near-parity observed in modern iterations. Overall, 73% of athletes in the dataset are male. Figure 2 shows a more dynamic view of gender participation in the Olympic games over time.

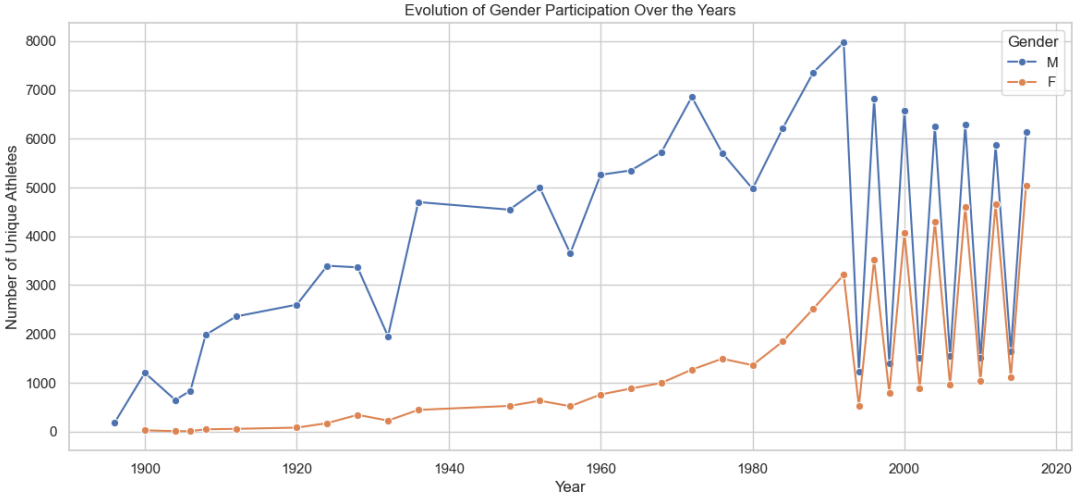

Figure 2: Temporal evolution of Olympic participation by gender. Note that the Summer and Winter Olympic Games were held in the same year until 1992 and have been staggered every two years since 1994.

3.3 Morphological Specialization

Utilizing biometric metrics (height and weight), the EDA identified a distinct polarization of physical traits, pointing out the specialization of athletes in the last years. A comparative study of biomechanically divergent disciplines—specifically Gymnastics, Basketball, and Weightlifting—reveals an increasing anthropometric distancing over the 80-year period since 1936.

3.4 Geopolitical impact

By identifying non-participation through a timeline-gap algorithm, the study shows the impact of Cold War-era boycotts and the exclusion of nations due to international sanctions. Furthermore, the dataset acts as a digital record of national obsolescence, tracking the final appearances of extinct supranational entities such as the Soviet Union and East Germany.

3.5 Home-Field Advantage

The research quantified the Host Nation Effect, correlating the role of the organizing country with a statistically significant spike in medal acquisition. By mapping host nations across 120 years, the EDA demonstrates a consistent performance improvement during the hosting year.

4 Related Work

4.1 Literature Review

The "120 years of Olympic history" dataset has inspired numerous analytical projects, providing a strong baseline for our research. Many existing works effectively establish the macro-historical context through comprehensive dashboards. For instance, the Tableau visualization by Caroline Zhu (Olympic Medal Analysis) aggregates historical data to highlight the all-time top medal-winning nations, offering a clear and essential snapshot of overall global dominance. Similarly, existing demographic analyses have effectively mapped out overarching trends in biometric averages and the total volume of gender participation across the century. Furthermore, developers have laid the groundwork for interactive data exploration; tools like Jsos17's web application (Olympics_Evolution) provide robust filtering systems that allow users

to retrieve specific medal statistics by year and country. Our project uses these established static baselines and functional search tools as a point of departure, aiming to expand upon their insights by injecting temporal dynamism and a narrative structure.

4.2 Originality of Our Approach

While existing projects treat the dataset as a fixed record, our approach conceptualizes it as a living timeline. Our originality relies on three core pillars:

- **Narrative-Driven Scrollytelling:** Instead of overwhelming users with complex multi-filter dashboards, we employ a curated narrative. We hook the audience with the relatable "Armchair Coach Syndrome," guiding them through the data to ensure insights are grounded in a human-centric context.
- **Time as the Engine:** We discard 120-year static aggregations. Utilizing animated charts and time sliders, we visualize active metamorphosis: how athlete morphology hyper-specializes over decades, how geopolitical maps dynamically redraw during the Cold War, and how sports disappeared.
- **The Discovery Factor:** By integrating Wikipedia metadata, features like the "Graveyard of Olympic Sports" elevate the project into an interactive discovery engine. Users actively uncover bizarre historical anomalies (e.g., Live Pigeon Shooting), effectively bridging quantitative data science with qualitative historical education.

References

Griffin, R. (2018a). 120 years of olympic history: Athletes and results, https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results. Accessed: 2026-03-18.

Griffin, R. (2018b). Discussion: 120 years of olympic history dataset, https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results/discussion. Kaggle discussion forum, accessed: 2026-03-18.